

Google EmbeddingGemma 2 Technical Report

1. Architecture Overview

EmbeddingGemma 2 maps text, code, images, audio, and video into a unified 768d space.

The base architecture utilizes an adapted Gemma 4 decoder with 8,192 token context.

The vision module adds 170M parameters to process images, PDFs, and video frames.

2. Matryoshka Representation Learning

Matryoshka Representation Learning enables dynamic vector truncation.

Embeddings can be truncated from 768d to 512d, 256d, or 128d.

Truncating to 128d achieves 6x storage reduction with over 90% accuracy retention.

The audio encoder adds 300M parameters for 16 kHz raw waveforms.